Week6
Median / order statistics

Week6 – On-site Examinations

- Regular1 - UVa 10931 – Parity [UVa 10931]
- Regular2 - UVa 10062 – Tell me the frequencies [UVa 10062]

- Bonus1 - UVa 10107 - What is the Median? [UVa 10107]
- Bonus2 - UVa 501 - Black Box [UVa 501]

UVa 10931 - Parity

- Integer representation

Decimal	Binary	# of 1s
0	00000	0
1	00001	1
2	00010	1
3	00011	2
4	00100	1
5	00101	2
8	01000	1
15	01111	4
16	10000	1

MSB							LSB
0	0	0	0	0	0	0	0
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Decimal ⇌ Binary

UVa 10062 – Tell me the frequencies

- Given a line of text you will have to find out the frequencies of the ASCII characters present in it.

```
     The decimal set:

       0 nul    1 soh    2 stx    3 etx    4 eot    5 enq    6 ack    7 bel
       8 bs     9 ht    10 nl    11 vt    12 np    13 cr    14 so    15 si
      16 dle   17 dc1   18 dc2   19 dc3   20 dc4   21 nak   22 syn   23 etb
      24 can   25 em    26 sub   27 esc   28 fs    29 gs    30 rs    31 us
      32 sp    33  !    34  "    35  #    36  $    37  %    38  &    39  '
      40  (    41  )    42  *    43  +    44  ,    45  -    46  .    47  /
      48  0    49  1    50  2    51  3    52  4    53  5    54  6    55  7
      56  8    57  9    58  :    59  ;    60  <    61  =    62  >    63  ?
      64  @    65  A    66  B    67  C    68  D    69  E    70  F    71  G
      72  H    73  I    74  J    75  K    76  L    77  M    78  N    79  O
      80  P    81  Q    82  R    83  S    84  T    85  U    86  V    87  W
      88  X    89  Y    90  Z    91  [    92  \    93  ]    94  ^    95  _
      96  `    97  a    98  b    99  c   100  d   101  e   102  f   103  g
     104  h   105  i   106  j   107  k   108  l   109  m   110  n   111  o
     112  p   113  q   114  r   115  s   116  t   117  u   118  v   119  w
     120  x   121  y   122  z   123  {   124  |   125  }   126  ~   127 del

FILES
     /usr/share/misc/ascii
```

Linux: man ascii

UVa 10062 – Tell me the frequencies

The **decimal** set:

0	nul	1	soh	2	stx	3	etx	4	eot	5	enq	6	ack	7	bel
8	bs	9	ht	10	nl	11	vt	12	np	13	cr	14	so	15	si
16	dle	17	dc1	18	dc2	19	dc3	20	dc4	21	nak	22	syn	23	etb
24	can	25	em	26	sub	27	esc	28	fs	29	gs	30	rs	31	us
32	sp	33	!	34	"	35	#	36	$	37	%	38	&	39	'
40	(	41	)	42	*	43	+	44	,	45	-	46	.	47	/
48	0	49	1	50	2	51	3	52	4	53	5	54	6	55	7
56	8	57	9	58	:	59	;	60	<	61	=	62	>	63	?
64	@	65	A	66	B	67	C	68	D	69	E	70	F	71	G
72	H	73	I	74	J	75	K	76	L	77	M	78	N	79	O
80	P	81	Q	82	R	83	S	84	T	85	U	86	V	87	W
88	X	89	Y	90	Z	91	[	92	\	93	]	94	^	95	_
96	`	97	a	98	b	99	c	100	d	101	e	102	f	103	g
104	h	105	i	106	j	107	k	108	l	109	m	110	n	111	o
112	p	113	q	114	r	115	s	116	t	117	u	118	v	119	w
120	x	121	y	122	z	123	{	124	|	125	}	126	~	127	del

FILES

/usr/share/misc/ascii

Sample Input

```
AAABBC
122333
```

Sample Output

```
67 1
66 2
65 3

49 1
50 2
51 3
```

The ascending order of their frequencies

Week6 – After Class Exercise

- LeetCode: 1122. Relative Sort Array [LeetCode 1122]
- LeetCode: 378. Kth Smallest Element in a Sorted Matrix [LeetCode 378]

Week7 - Before Class Exercise

- LeetCode: 704. Binary Search [LeetCode 704]
- LeetCode: 74. Search a 2D Matrix [LeetCode 74]